

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Six Semester of B. Tech. (EE) Examination

April-May 2013

EE309 Electric Power Utilization & Traction

Date: 03.05.2013, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Explain plugging in d.c. series and shunt motor with torque equations. [07]

Q - 2 (a) Write speed control methods of three phase induction motor and explain any three. [07]

(b) What is electric drive Explain types of Drive and write advantages of electric drive. [07]

OR

Q - 2 Write Short notes: (Any Two) [14]

(1) Track electrification.

(2) photometry.

(3) Speed control of d.c. series motor.

Q - 3 (a) What is Traction system? Give classification of Traction system. And write features of an ideal traction system. [07]

(b) The speed time curve of a train consists of: [07]

(i) Uniform acceleration of 6 kmphs for 25 seconds;

(ii) Free running for 10 minutes;

(iii) Uniform deceleration of 6 kmphs to stop the train;

(iv) A stop of 5 minutes;

Find the distance between the stations, the average and schedule speeds.

OR

Q - 3 (a) Draw simplified speed time curve and Calculate V₂ by quadrilateral speed time curve. [07]

(b) An electric train has quadrilateral speed-time curve as follows. [07]

(i) uniform acceleration from rest at 2 kmphs for 30 seconds. (ii) coasting for 50 seconds. (iii) braking period of 20 seconds.

The train is moving a uniform up gradient of 1 % tractive resistance is 40 newtons per tonne, rotational inertia effect 10% of dead weight, duration of station stop 15 seconds

and overall efficiency of transmission gear and motor as 75% .calculate the value of its schedule speed and specific energy consumption of run.

SECTION – II

- Q - 4 (a) Give Comparison between high frequency & power frequency heating. [07]
- Q - 5 (a) Write the Comparison between tungsten Filament lamps and Fluorescent tubes. [06]
- (b) Explain basic Principal of electrolysis. [03]
- (c) Write a short note on Direct & Indirect Arc furnace. [05]

OR

- Q - 5 (a) State and Explain laws of illumination. [06]
- (b) State and Explain Faraday's law of Electrolysis. [04]
- (c) Explain Factory Lighting. [04]
- Q - 6 (a) Derive flywheel calculation equation when motor load is increasing and motor load is decreasing. [07]
- (b) A 4 pole, 50Hz induction motor has a flywheel on its shaft. Total inertia at the motor shaft is 1000 kg-m^2 . The load torque is 100 kg-m for 10 second followed by a no load period long enough for the flywheel to regain its full speed. Motor has a slip of 6% at a torque of 50 kg-m . Calculate the speed at the end of deceleration period. Assume motor speed-torque characteristic to be a straight line in the region of interest & neglect friction and windage loss. [07]

OR

- Q - 6 (a) Derive Equation for Temperature rise of electric drive. (Heating curve and cooling curve) [07]
- (b) A certain motor has to perform the following duty cycle [03]
- 100 h.p. for 10 minutes, No load for 5 minutes, 50 h.p. for 8 minutes, No load for 4 minutes. The duty cycle is repeated indefinitely. Draw the curve for the load cycle. Assuming that the heating is proportional to square of the load, determine the suitable size of a continuously rated motor.
- (c) Write Short Note on Spot Welding. [04]
